Labreport Advanced Computer Architecture
Lab 5

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March 25, 2024



Class: MEIsw

Contents

1 Introduction **1**

2 Synchronous vs asynchronous execution **1**

1 Introduction

In this lab session we explored synchronous and asynchronous execution. The source code for this project can be found on
`https://github.com/RoelsJonas/Advanced-Computer-Architecture-Lab-5`

2 Synchronous vs asynchronous execution

For this experiment we wrote two programs, one which executes the four different array operation sequentially and another one which executes in parallel. To test with bigger array sizes we also implemented striding in every operation. Thus for our experiments we used a array size of 69212 which made the ratio of CPU runtime vs GPU runtime smaller and made it easier to spot any advantages between the sequential and parallel methods. Looking at the latter we can see that this runs slightly faster than the sequential approach.

Note that to create a single array the CPU needs on average 3.53 ms. Thus to create the four arrays the CPU would on average need 14.12 ms. Therefore the result of 16.39 ms on average for the parallel execution means that there is only a little bit of overhead due to the copying of memory to the GPU. The sequential execution had an average processing time of 21,15 ms. In Figure 1 we see a visual representation of these results.

An important remark here is that the results of this experiment are highly dependant on the relative time required by the GPU and CPU functions. The parallel execution will have the biggest advantage if these are of comparable length.

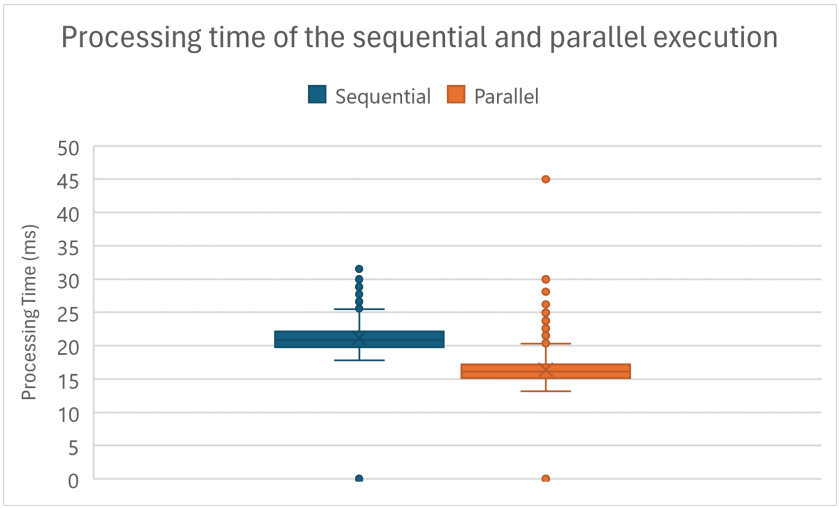


Figure 1: Comparison of the processing time between sequential and parallel execution